CESIM TEACHING NOTE

Design Thinking Microsimulation

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1. Simulation Overview

This simulation offers a practical, hands-on introduction to the design thinking process, guiding participants to develop solutions that align with user needs, business goals, and technical feasibility. Set in a dynamic, competitive market scenario, the challenge emphasizes solving real-world problems where consumer expectations are rapidly evolving.

Participants will move through the five phases of design thinking—Empathize, Define, Ideate, Prototype, and Test—applying insights and creativity to create user-centered innovations. By the end of the experience, they will gain a solid understanding of design thinking, improve strategic decision-making under constraints, and learn to balance desirability, feasibility, and viability in product development.

Learning Objectives

Upon completion of this simulation, participants will be able to:

- Apply the design thinking methodology (Empathize, Define, Ideate, Prototype, Execute) to develop user-centered products
- Make data-driven decisions by analyzing market research, user sentiment, and competitive landscape
- Balance multiple constraints including budget, time, technical feasibility, and user requirements
- Optimize product features based on quantitative parameters that affect market attractiveness and profitability
- Develop strategic thinking by aligning product development with market opportunities and user needs
- Experience the iterative nature of design thinking through multiple simulation rounds
- Navigate cross-functional perspectives and priorities in the product development process

Target Audience

- Graduate and undergraduate students in business, engineering, design, and product management programs
- Product managers and professionals seeking to enhance their design thinking capabilities
- Entrepreneurs developing new products for specific market segments
- Cross-functional teams working on innovation initiatives

Time Required

- Preparation: 30 minutes
- Simulation gameplay: 60-90 minutes (2-3 rounds)
- Debrief discussion: 45-60 minutes
- Real-world application activity: 30 minutes
- Total: 3-3.5 hours

Case Scenario and Winning Criteria

Participants take on the role of a product development team at a company with ten years of experience in delivering innovative consumer products and technology solutions. Now entering a new product category, the company aims to develop offerings with localized features tailored to specific market segments. In the simulation, participants make strategic decisions that shape the success of their product. The end goal is to design a product that resonates with users and performs strongly in the market, by effectively balancing user desirability, technical feasibility, and business viability. Specifically, participants are expected to increase market share by 2% and achieve a profit margin of 8% through their strategic choices.

Simulation Structure

The microsimulation is organized into independent rounds, each presenting either consistent scenarios or allowing the instructor to customize market conditions to simulate decision-making challenges throughout the company's growth journey.

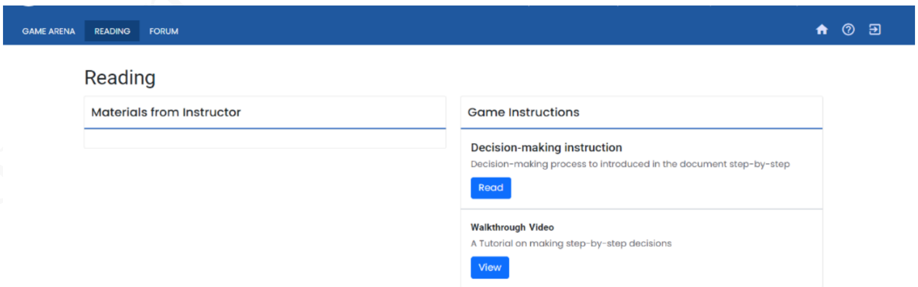
The game arena is designed that including the five stages of design thinking:

1. **Observe**: Analysing market data and trends to identify opportunities
2. **Empathize**: Understanding user personas, journey maps, and sentiment analysis to identify needs
3. **Define**: Identifying market gaps and constraints to formulate the design challenge
4. **Ideate**: Generating product concepts based on user needs and market opportunities
5. **Prototype**: Developing and testing product features with varying levels of fidelity
6. **Execute**: Launching the product and analysing performance metrics

Simulation Structure

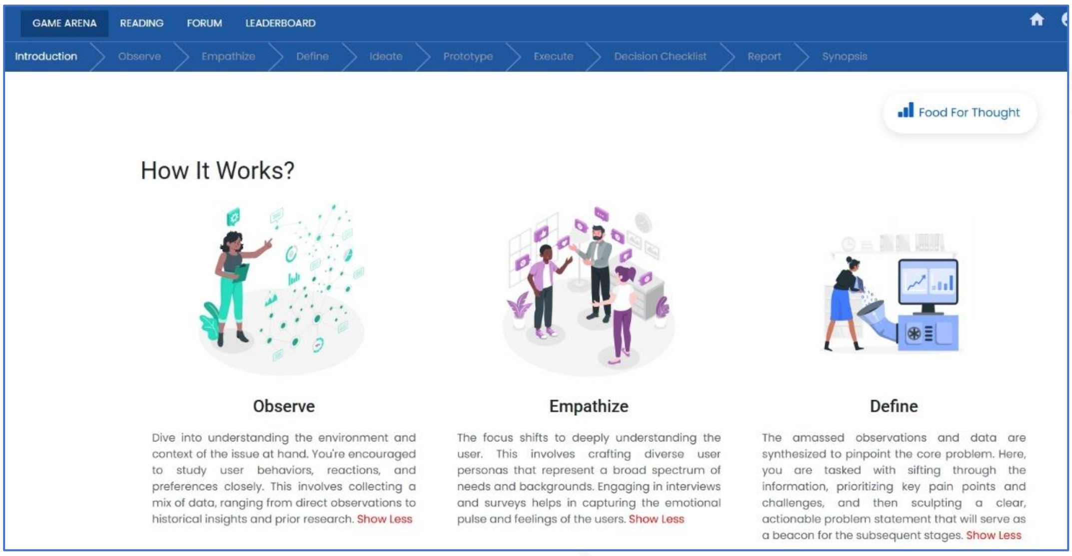
2. Pre - Simulation Preparation

a. **Review Background Materials**



- Login as participant to explore the platform
- Read the decision-making guide thoroughly
- Watch the walkthrough video for practical insights
- Understand the company context and business objectives

b. **Familiarize with the Interface**

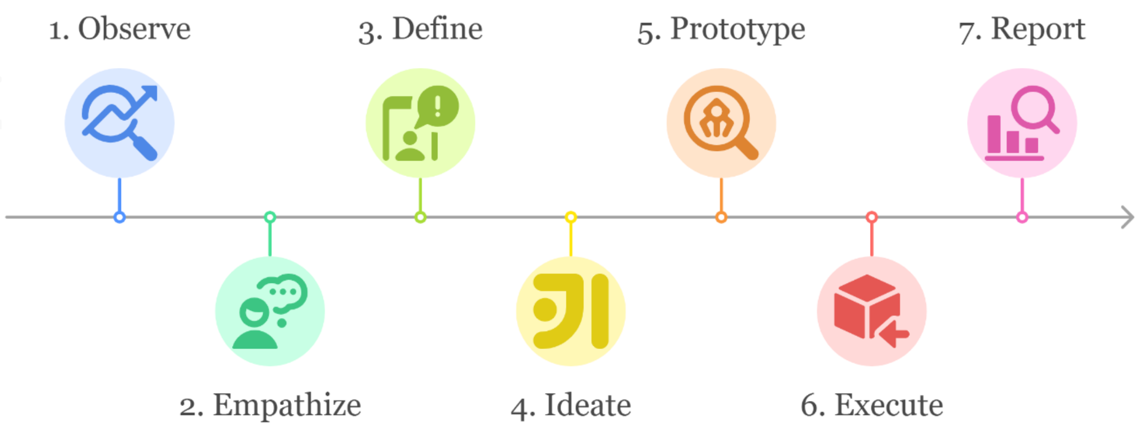


- Explore the Game Arena layout
- Understand the sequence of decision tabs
- Review the "Food for Thought" section for critical considerations

c. **Understanding Decision-Making Process**

The simulation follows a logical step-by-step process. Participants should move through each stage to make thoughtful and informed decisions.

Decision Making Flow



The simulation utilizes a complex algorithm that calculates:

1. **Attractiveness Score**: Based on feature selection, localization, and warranty period
   - Formula: Base Attractiveness + (Feature Impact × Feature Selection) + (Warranty Multiplier × Warranty Period) + (Localization Impact × Localization %)
   - Theoretical basis: Kano model of customer satisfaction, conjoint analysis
2. **Market Share**: Derived from attractiveness score, price positioning, and channel effectiveness
   - Formula: Base Market × (Attractiveness Elasticity × Attractiveness Score) × (Price Elasticity × Price Factor) × (Channel Investment Effectiveness)
   - Theoretical basis: Diffusion of innovation theory, marketing mix theory
3. **Operating Margin**: Calculated from revenue, variable costs, and fixed costs
   - Formula: (Revenue - Variable Cost - Fixed Cost) / Revenue
   - Theoretical basis: Value-based pricing, cost structure analysis
4. **Unit Sales**: Driven by market size, market share, and product attractiveness
   - Formula: Available Market × Market Share × (1 + Attractiveness Modifier)
   - Theoretical basis: Market penetration theory, product adoption cycles

These calculations create a system of interdependent variables that reflect the complex dynamics of real-world product development.

Key instructor-only parameters include:

These back-end parameters are critical for instructors to understand the simulation mechanics:

1. **Elasticity Curves**:
   - Price elasticity (0.1 to 3.0) based on attractiveness score
   - Promotional elasticity (0.68 to 1.18) based on spending
   - Battery life elasticity (1.0 to 2.4) based on battery performance
   - Feature elasticity impacts on attractiveness:
     - Design & Aesthetics: 1.04
     - Localized Language Support: 1.06
     - Unique Health Features: 1.04
     - Indian Payment Integration: 1.05

2. **Channel Effectiveness**:
   - Year 1-5 performance trends by channel
   - E-commerce effectiveness increases over time (70 to 88)
   - Distributor effectiveness decreases over time (67.5 to 61)

3. **Competitor Benchmarks**:
   - Global Giant promotion budget trajectory (336M to 1680M)
   - Mid-sized international promotion trends (134M to 403M)
   - Local Player benchmarks (67M to 268M)

4. **Base Parameters for Product Concepts**:
   - TechTime Vital+: Attractiveness (82), Battery Life (1.1 days), Localization (20%), Base Cost (14000)
   - TechTime PayBharat: Attractiveness (84), Battery Life (0.5 days), Localization (29%), Base Cost (12000)
   - TechTime KidGuard: Attractiveness (81), Battery Life (0.8 days), Localization (24%), Base Cost (13000)

5. **Market Size Parameters**:
   - Total Population: 1,400 million
   - Total Addressable Market: 56,000
   - Serviceable Available Market: 5,500

Teaching with this Information

During debriefing, use these parameters to explain why certain student strategies succeeded or failed. This transforms the simulation from a "black box" into a transparent demonstration of Design Thinking Principles.

Common Participant Challenges

Anticipate these common issues:

1. **Feature Overload**: Participants often try to maximize all features without considering cost implications.
   - Intervention: Ask them to rank features based on user data rather than personal preferences.
2. **Price Fixation**: Teams may focus too much on pricing without adequately developing product attractiveness.
   - Intervention: Prompt them to reconsider their value proposition before finalizing pricing.
3. **Ignoring User Data**: Some participants jump to solutions without thoroughly analysing user personas and sentiment data.
   - Intervention: Ask specific questions about user needs that require returning to the empathy data.
4. **Channel Misalignment**: Investments in channels may not align with the target user segments.
   - Intervention: Have teams explicitly connect their channel strategy to user journey maps.
5. **Linear Thinking**: Treating the design process as sequential rather than iterative.
   - Intervention: Encourage revisiting earlier phases when new information emerges.

Final Checklist Before Introducing the Simulation

- Complete the simulation yourself to understand decision flows and core mechanics.
- Familiarize yourself with instructor-only insights, including critical mechanics and performance indicators.
- Prepare examples of how to interpret data from each simulation phase
- Ensure all participants are registered using the simulation link and have successfully enrolled before the session.
- Coordinate with IT and set up necessary infrastructure in advance to ensure smooth implementation and avoid disruptions.

3. Facilitation Roadmap

Pre-Simulation Briefing

1. **Introduction**
    - Explain the design thinking methodology and its application in product development
    - Introduce TechTime Innovations and the wearable technology market context
    - Outline the simulation objectives and winning criteria
    - Connect the simulation to broader design thinking theory and practice

2. **Interface Walkthrough**
    - Demonstrate the navigation between design thinking phases
    - Explain the difference between information screens and decision points
    - Show how to interpret data visualizations and market research
    - Highlight the feedback mechanisms that connect decisions to outcomes
    - Demonstrate the decision consequence visualization tools

3. **Decision Framework**
    - Explain the key decisions participants will make in each phase
    - Highlight the interdependencies between decisions
    - Discuss how to balance user needs, technical feasibility, and business viability
    - Explain team roles if using the cross-functional team structure
    - Set expectations for team collaboration and participation

During Simulation

Observe

- Encourage thorough analysis of market data and trends
    - Prompt: "What patterns do you see in the market data? How might these influence your approach?"
    - Theory connection: "Consider how market sensing theory applies here."

Empathize

- Prompt detailed examination of user personas and journey maps
    - Prompt: "What unstated needs can you identify from the personas? How do the journey maps reveal pain points?"
    - Theory connection: "How does this relate to jobs-to-be-done theory?"

Define

- Guide participants to identify the most significant gaps and constraints
    - Prompt: "Based on your user research, what is the core problem you're solving?"
    - Theory connection: "Consider how problem framing affects solution possibilities."

Ideate

- Facilitate creative thinking while maintaining feasibility
    - Prompt: "How might your features address the specific needs identified in your empathy work?"
    - Theory connection: "Notice how divergent thinking expands your solution space."

Prototype and Execute

To guide participants in reflecting on how their **feature and prototype decisions** translated into **marketing strategy choices** during execution, and how these influenced performance metrics like attractiveness, market share, and profitability.

1. Feature Prioritization & Assumptions Testing

- **Prompt:** "What key user needs or assumptions did each prototype aim to address?"
- **Prompt:** "How did unit cost and feature combinations influence which prototype you selected for execution?"
- **Theory Connection:** MVP mindset, risk mitigation, and Jobs-to-be-Done theory.

2. Decision Criteria for Execution Selection

- **Prompt:** "What criteria did you use to select the prototype that moved into execution?"
- **Prompt:** "Did your selection optimize cost-efficiency, innovation, or desirability?"
- **Theory Connection:** Desirability–Feasibility–Viability lens, user-centered design.

Execution Phase

Participants moved into **strategic marketing decisions** (Price, Advertising, Warranty, Channel Investment) based on the prototype they selected. The debrief should focus on how well they aligned marketing tactics with the product's value proposition.

1. Alignment Between Product Features and Marketing Mix

- **Prompt:** "How did your feature choices influence your marketing mix decisions?"
- **Prompt:** "What story were you telling in the market through your pricing and channel investments?"
- **Theory Connection:**
  ➤ Product-Market Fit, 4Ps of Marketing (Product, Price, Place, Promotion).

2. Strategic Trade-Offs

- **Prompt:** "What were the most difficult trade-offs you faced when allocating your marketing budget?"
- **Prompt:** "How did you balance short-term reach vs. long-term brand positioning?"
- **Theory Connection:**
  ➤ Customer Lifetime Value, marketing elasticity, value-based pricing.

3. Anticipating Performance Without Immediate Feedback

- **Prompt:** "Without seeing performance feedback during decision-making, how did you evaluate the potential impact of your execution choices?"
- **Prompt:** "What indicators or data did you rely on to justify your mix?"
- **Theory Connection:**
  ➤ Strategic foresight, scenario planning, decision-making *under uncertainty*.

2. **Between Rounds**
   - Facilitate brief team discussions about results and learnings
     - Prompt: "What surprised you about your results? Which decisions had the greatest impact?"
   - Encourage teams to identify what worked and what didn't
     - Prompt: "Which assumptions proved correct? Which need revisiting?"
   - Prompt teams to consider how they'll adjust their strategy
     - Prompt: "How will you iterate your approach based on these learnings?"
   - Monitor team dynamics and intervene if participation is unbalanced
     - Intervention: "Let's hear from each team member about one insight from this round."

Debrief Structure

After execution, participants view their **performance scores** (Attractiveness, Market Share, Operating Margin). Use this to close the loop:

1. Connecting Decisions to Outcomes

- **Prompt:** "Looking at your report, which marketing decision do you think had the biggest impact on your results?"
- **Prompt:** "If your results were below expectations, which earlier assumptions might have been flawed?"

2. Iterative Learning for Future Rounds

- **Prompt:** “What would you test or do differently in the next iteration?”
- **Prompt:** “How can you better integrate user insights into future marketing decisions?”
- Use comparison across teams to discuss **divergent strategies** and **differing results**.
- Have teams reverse-engineer top-performing teams' **execution logic**.
- Emphasize the **feedback loop** from outcome → reflection → revised strategy.

1. **Results Analysis** (
    - Display final leaderboard and key performance metrics
    - Have top-performing teams briefly explain their approach
    - Highlight different strategies that led to success
    - Analyze failure scenarios that illustrate common pitfalls
        - Example: "Team C's focus on features without user validation led to high costs but low market share."
    - Use decision consequence visualizations to show how early decisions shaped outcomes

2. **Decision Process Review**
    - Examine key decision points and their impacts
    - Discuss how teams balanced competing priorities
    - Analyze how market insights influenced feature selection
    - Review team dynamics and collaboration effectiveness
        - Prompt: "How did your team make decisions? What worked well or could be improved?"
    - Connect decision processes to design thinking phases
        - Prompt: "Which phase did your team excel at? Which was most challenging?"

3. **Design Thinking Application**
    - Evaluate how effectively teams applied each phase of design thinking
    - Identify which teams most thoroughly empathized with users
    - Discuss the value of iteration in the design process
    - Analyze how prototype fidelity choices affected outcomes
    - Explicitly connect simulation experiences to theoretical frameworks
        - Example: "Team A's approach illustrates the importance of divergent thinking before convergent decision-making."

4. **Real-World Connections**
    - Link simulation decisions to actual product development challenges
    - Discuss how these principles apply across different industries
    - Connect to broader business strategy and innovation concepts
    - Invite participants to share similar challenges from their experiences
    - Introduce the real-world application activit

5. **Real-World Application Activity**
   - Have participants identify a real challenge in their organization or studies
   - Apply the design thinking process to develop an initial approach
   - Create a one-page plan outlining:
     - Key user needs and insights (Empathize)
     - Problem definition (Define)
     - Potential solutions (Ideate)
     - Testing approach (Prototype)
     - Implementation considerations (Execute)
   - Share plans in small groups for feedback
   - Discuss enablers and barriers to applying design thinking in their contexts

4. Critical Decision Points Analysis

Decision Point 1: Market Segmentation & User Needs Prioritization

Key Decision: Determining which user needs to prioritize based on market research, personas, and sentiment analysis.

Optimal Approach

- Thoroughly analyse all user personas to identify common pain points
- Cross-reference sentiment analysis (65% positive, 20% neutral, 15% negative) with feature priorities
- Prioritize features with high preference percentages (Battery Life: 85%, Design & Aesthetics: 70%, Affordability: 65%)
- Identify gap between current market offerings and user expectations (e.g., battery life of 1.5 days vs. desired 3.5 days)
- Target segments with strongest growth potential (health-conscious urban adults)
- Apply jobs-to-be-done theory to understand the functional, emotional, and social dimensions of user needs

Common Pitfalls

- Skimming user research rather than deeply analyzing needs and motivations
- Focusing on a single persona rather than finding common patterns across user types
- Neglecting the sentiment analysis data that reveals pain points
- Missing the gap analysis that shows where current products fall short
- Prioritizing features based on team preferences rather than user data
- Failure to consider the cultural context of user needs

Decision Impact Matrix

User Need	Market Gap	User Preference (%)	Impact on Attractiveness	Optimal Priority	Theoretical Connection
Battery Life	2-day gap (1.5 vs. 3.5)	85%	1.05 multiplier at 100%	Very High	Performance need in Kano model
Localization	40% gap (20% vs. 60%)	60%	1.06 multiplier	High	Cultural context in UX design theory
Health Features	Limited unique offerings	50%	1.04 multiplier	Medium	Jobs-to-be-done: functional dimension
Payment Integration	Limited availability	45%	1.05 multiplier	Medium-High	Ecosystem theory in product design
Affordability	₹12,000 gap between tiers	65%	Price elasticity 0.6-3.0	High	Value perception theory

Decision Point 2: Feature Selection & Optimization

Key Decision: Selecting and balancing product features to maximize attractiveness while managing technical feasibility and cost.

Optimal Approach

- Analyze the elasticity curves for each feature to determine ROI on feature investment
- Optimize battery life based on the high elasticity (1.0-2.4) and strong preference (85%)
- Balance feature investment based on cost multipliers (higher for localized features at 1.13)
- Prioritize features with stronger competence elasticity (Localized Language: 1.09, Payment Integration: 1.09)
- Adjust warranty period based on attractiveness impact (12 months: 75 points, 24 months: 90 points)
- Consider feature combinations that create coherent user experiences rather than isolated functionalities
- Apply Kano model to distinguish between basic needs, performance needs, and delight features

Common Pitfalls

- Attempting to maximize all features without considering cost implications
- Neglecting high-elasticity features that deliver greater returns on investment
- Overinvesting in features with limited impact on target segments
- Selecting inconsistent features that don't create a coherent product story
- Ignoring technical dependencies between features
- Focusing exclusively on functional features while neglecting emotional and social dimensions

Feature Optimization Table

Feature	Base Cost	Cost Multiplier	Attractiveness Multiplier	Competence Elasticity	Optimal Investment	Kano Model Classification
Battery Life	Varies by %	1.02-1.29	0.5-1.05	1.0-1.15	65-75%	Performance need
Localized Language	Varies	1.13	1.06	1.09	Active	Excitement feature
Water Resistance	Varies	1.09	1.025	1.035	Active	Basic need for specific segments
Health Features	Varies	1.12	1.04	1.07	Active	Performance/excitement feature
Payment Integration	Varies	1.15	1.05	1.09	Active for urban segment	Excitement feature

Decision Point 3: Prototype Development & Testing Approach

Key Decision: Determining the appropriate level of prototype fidelity and testing methodology to validate key assumptions.

Optimal Approach

- Assess critical assumptions that need validation before full development
- Align testing methodology with target user segments
- Prioritize testing of high-impact features identified in earlier phases
- Plan for incorporating feedback into subsequent iterations
- Structure testing to generate both qualitative insights and quantitative metrics

Common Pitfalls

- Designing tests that confirm biases rather than challenge assumptions
- Failure to capture both emotional and functional user responses
- Inadequate planning for feedback integration
- Rushing through testing to move to execution

Prototype Testing Strategy Matrix

Assumption Type	Risk Level	Recommended Fidelity	Testing Approach	Feedback Integration Plan
Core Value Proposition	High	Medium to High	In-depth interviews with 5-8 target users	Revisit problem definition if feedback negative
Feature Usability	Medium	Medium	Task completion testing with 8-12 users	Iterate features based on usability metrics
Aesthetic Appeal	Medium	High	A/B testing with larger sample	Adjust design elements based on preference data
Price Sensitivity	High	Low to Medium	Simulated purchase scenarios	Refine pricing strategy based on willingness-to-pay
Channel Preference	Medium	Low	User journey mapping exercises	Adjust channel investment based on preference patterns

Decision Point 4: Pricing & Channel Strategy

Key Decision: Determining optimal price point and distribution channel mix to maximize market reach and profitability.

Optimal Approach

- Set price based on elasticity curve that shows optimal pricing around ₹27,000 for premium features
- Analyze channel effectiveness data showing e-commerce growing from 70 to 88 over 5 years
- Allocate channel investment with emphasis on e-commerce (₹200M) and telecom partnerships (₹150M)
- Consider the declining effectiveness of distributor/wholesaler channels (67.5 to 61)

- Balance advertising investment (₹350M) against promotional elasticity curve (1.15 at ₹350M)
- Align pricing and channel strategy with user personas and journey maps
- Apply value-based pricing principles that reflect the product's benefits to users

Common Pitfalls

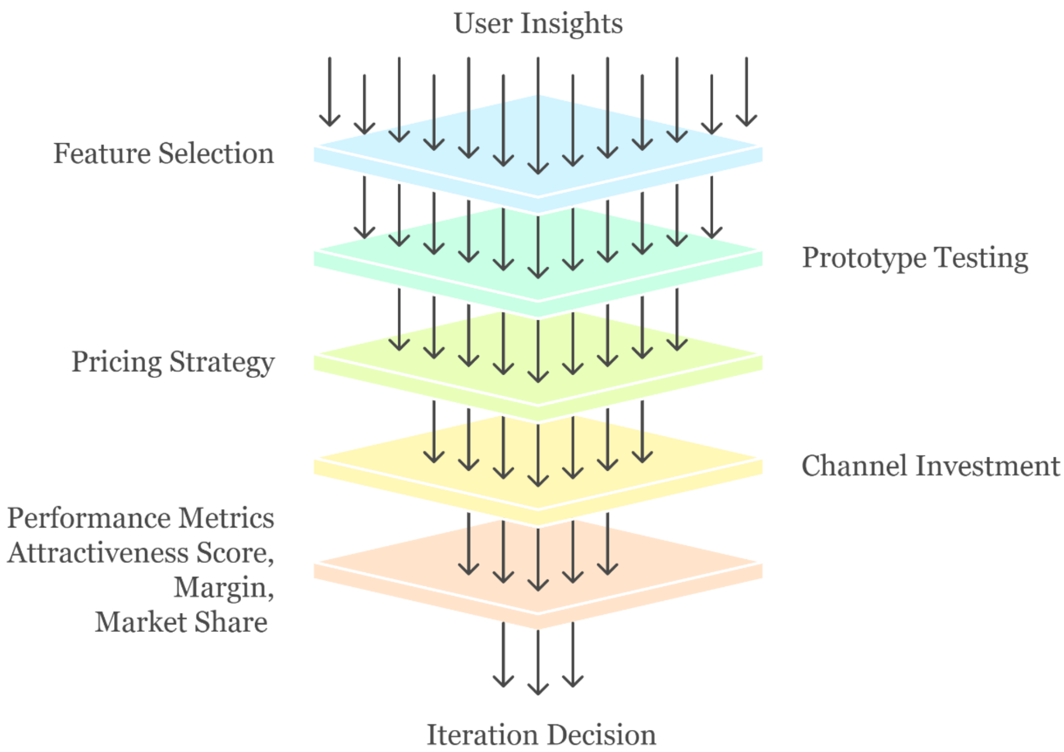
- Pricing too low in an attempt to gain market share without considering margin impact
- Under-investing in growing channels like e-commerce
- Allocating equal investment across all channels rather than prioritizing
- Setting promotion budget without considering competitive benchmarks
- Disconnecting pricing strategy from the value proposition established in earlier phases
- Ignoring how channels influence the overall user experience

Channel Strategy Table

Channel	Year 1 Effectiveness	Year 5 Trend	Investment Multiplier	Optimal Investment (INR Million)	Key User Segments	Journey Map Touchpoints
E-commerce	70	88 (↑)	1.07 at ₹200M	200	Tech-savvy professionals	Research, comparison, purchase
Telecom Partnerships	60	70 (↑)	1.05 at ₹200M	150	Urban professionals	Discovery, trial, support
Distributor/Wholesaler	67.5	61 (↓)	1.04 at ₹200M	100	Traditional consumers	Physical examination, purchase
Direct-to-consumer	50	65 (↑)	1.03 at ₹200M	100	Brand loyalists	Research, purchase, advocacy

5. Understanding Decision Linkages and System Dynamics

The simulation requires participants to make decisions across five key areas, each connected to specific design thinking principles:



1. **Product Features**: Select and prioritize features based on user research (Empathize → Define)
    - Battery life: Balance user needs with technical constraints
    - Localization percentage: Address cultural relevance
    - Warranty period: Manage risk perception and trust
    - Design and aesthetics: Consider emotional connection
    - Language support: Enhance accessibility
    - Water resistance: Address usage context
    - Health features: Solve specific user problems
    - Payment system integration: Create ecosystem connections

2. **Pricing Strategy**: Set pricing based on value perception and market positioning (Define → Ideate)
    - Balance between affordability and premium positioning
    - Consider price elasticity and margin requirements
    - Align pricing with the value proposition

3. **Distribution Channels**: Allocate investment across different channels (Ideate → Prototype)
    - Distributor/Wholesaler: Reach traditional retail environments
    - E-commerce: Access digital consumers
    - Telecom Partnerships: Leverage existing relationships
    - Direct-to-consumer: Build direct customer connections

4. **Marketing**: Determine advertising budget and promotional strategy (Prototype → Execute)
    - Consider promotional elasticity
    - Compare against competitor spending
    - Align messaging with user needs identified in the empathy phase

5. **Product Portfolio**: Select which product concepts to develop and launch (Execute → Iterate)
    - TechTime Vital+ (health-focused)
    - TechTime Yatri (travel-focused)
    - Additional specialized concepts based on market segments

Summary of Core Decision Areas & Key Parameters

Decision Area	Key Parameters	Impact on KPIs	Optimal Approach
Feature Selection	Feature elasticities (1.01-1.06)	Drives attractiveness score	Balance high-impact features with costs
Prototype Testing	Fidelity levels, feedback types	Validates assumptions	Match fidelity to risk level
Product Pricing	Price elasticity (0.6-3.0)	Affects volume and revenue	Price based on attractiveness and target segment
Channel Strategy	Channel effectiveness (50-88)	Influences market reach	Prioritize growing channels like e-commerce
Promotion	Promotional elasticity (0.68-1.18)	Impacts awareness and share	Invest based on competitive benchmarks
Product Portfolio	Base parameters per concept	Determines market coverage	Select complementary concepts for segments

6. Performance Analysis Framework

After each round, participants should review the results in the **Report** and **Synopsis** tabs to assess the impact of their decisions. This process encourages reflection on what worked well, what didn't, and how to improve in the next round. It's not just about understanding what happened—but why it happened.

Attractiveness Score Analysis

The Attractiveness Score represents how appealing the product is to target users and directly influences market share.

Key Components:

- Base product concept attractiveness (70-84 points)
- Feature selection impact (multipliers of 1.01-1.06)
- Warranty period contribution (6 months: 60 points to 36+ months: 93 points)
- Localization percentage (20-29% base, with user demand at 60%)
- Battery life optimization (current: 1.5 days, target: 2.0 days)

Report Interpretation:

- Scores below 70 indicate suboptimal feature selection
- Scores 70-80 show adequate feature alignment with user needs
- Scores above 80 demonstrate excellent feature optimization
- The warranty period vs. attractiveness chart shows diminishing returns beyond 24 months

Decision Impact:

- Each 10-point increase in attractiveness can yield a 0.2-0.5% increase in market share
- Feature combinations with complementary effects (health features + localization) create synergies
- Battery life improvements offer the highest ROI on attractiveness (elasticity up to 2.4)

Market Share Analysis

Market share represents the percentage of the serviceable available market captured by your product.

Key Components:

- Attractiveness score elasticity (0.1% at 0 points to 3.0% at 100 points)
- Price positioning effect (elasticity from 0.6 to 3.0)
- Channel effectiveness and investment levels
- Competitive positioning against established players (Apple: 30%, Samsung: 20%)

Report Interpretation:

- Initial market share below 1% indicates weak market positioning
- Share of 1-2% represents moderate success
- Share above 2% indicates strong market penetration
- Track share changes between rounds to gauge strategy effectiveness

Decision Impact:

- E-commerce channel investment offers highest long-term return (effectiveness to 88)
- Premium pricing can work with high attractiveness scores (>80)
- Localized features can capture underserved market segments

Operating Margin Analysis

Operating margin measures profitability as a percentage of revenue.

Key Components:

- Revenue (price × units sold)
- Variable costs (influenced by feature selections)
- Fixed costs (production: ₹350M, administration: ₹150M, research: ₹60M)
- Advertising and channel investments

Report Interpretation:

- Negative margins indicate unsustainable pricing or excessive costs
- Margins of 0-5% suggest adequate but not optimal pricing
- Margins above 5% demonstrate strong price-value positioning
- Income statement breakdown shows contribution of each cost category

Decision Impact:

- Each feature addition increases unit cost through multipliers (1.02-1.29)
- Warranty extension increases costs (multipliers 1.02-1.08)
- Higher battery life significantly impacts cost (up to 1.29 multiplier at 100%)

Theoretical Connection:

- Value-based pricing (setting price based on perceived value)
- Cost structure analysis (fixed vs. variable costs)
- Contribution margin approach (per-unit profitability)

Units Sold Analysis

Units sold represents actual sales volume based on market share and available market.

Key Components:

- Serviceable available market size (5,500 million)
- Achieved market share percentage
- Attractiveness score multiplier effect

Report Interpretation:

- Units below 500,000 indicate limited market penetration
- 500,000-1,000,000 units represent moderate success
- Over 1,000,000 units show strong market acceptance
- Population breakdown shows total addressable market potential

Decision Impact:

- Channel investment affects reach into available market
- Price positioning influences volume through elasticity
- Promotional investment affects awareness and consideration

Summary of Key Performance Indicators

The leaderboard ranking is based on the Operating Margin

KPI	Target Range	Primary Drivers	Report Location	Interpretation	Theoretical Framework
Attractiveness Score	75-85	Feature selection, warranty, battery life	KPI section	Product appeal to target users	Kano model, conjoint analysis
Market Share	1.5-2.5%	Attractiveness, price, channels	KPI section	Portion of available market captured	Diffusion of innovation, blue ocean strategy
Operating Margin	4-8%	Price, costs, features	Income Statement	Profitability of operations	Value-based pricing, cost structure analysis
Units Sold	800K-1.2M	Market share, market size	KPI section	Actual sales volume	Market penetration, price-demand relationship

7. Theoretical Connections by Simulation Section

Decision Section	Simulation Area	Theory	Summary
Observe	Market Analysis	Market Sensing Theory	Gathering external data to identify trends and opportunities
	Competitive Analysis	Competitive Positioning Theory	Understanding how competitors are positioned and identifying gaps
Empathize	User Personas	Jobs-to-be-Done Theory	Understanding what "jobs" users are hiring the product to perform
	Journey Maps	Customer Experience Design	Mapping the customer journey from discovery to advocacy
	Sentiment Analysis	Voice of Customer	Analysing positive (65%), neutral (20%), and negative (15%) feedback

Define	Gap Analysis	Blue Ocean Strategy	Identifying uncontested market space
	Constraints	Resource-Based View	Recognizing limitations in resources and capabilities
	SWOT Analysis	Strategic Management	Evaluating internal and external factors
Ideate	Product Concepts	Divergent Thinking	Generating multiple ideas without immediate judgment
	Feature Prioritization	Kano Model	Categorizing features as basic, performance, or excitement
Prototype	Feature Selection	Conjoint Analysis	Evaluating feature combinations for overall value
	Prototype Fidelity	Usability Engineering	Balancing comprehensiveness with speed and cost
	User Testing	Build-Measure-Learn Cycle	Creating testable prototypes to gather feedback
Execute	Pricing Strategy	Value-Based Pricing	Setting price based on perceived value
	Channel Strategy	Omnichannel Distribution	Creating integrated customer experiences across channels
	Marketing Mix	4Ps Marketing Theory	Balancing product, price, place, and promotion
	Go-to-Market	Product Launch Strategy	Planning the introduction of new products
Iteration	Performance Analysis	Double-Loop Learning	Questioning assumptions based on results
	Pivot or persevere	Lean Startup Methodology	Deciding whether to refine or fundamentally change

8. Sample Optimal Solution

It is essential to acknowledge that numerous combinations can lead participants to achieve the desired outcomes, and the provided solution serves as a reference for faculty and is not intended to be shared directly with participants.

Table Summarizing the key decision areas of the Microsimulation

Decision Area	**Input**
Target: Battery Life, Days	2
Target: Affordability, INR	20,000
Target: Localization, %	40%
Ideate	
Idea 1	TechTime Linguist
Idea 2	TechTime PayBharat

Idea 3	TechTime Academia
Prototype & Test	
Product 1	TechTime Linguist
Battery Life	80%
Design & Aesthetics	No
Localized Language Support	No
Seamless Syncing	No
Water Resistance	Yes
Customizable Watch Faces	Yes
Unique Health Features	Yes
Indian Payment System Integration	No
Product 2	TechTime PayBharat
Battery Life	70%
Design & Aesthetics	Yes
Localized Language Support	Yes
Seamless Syncing	Yes
Water Resistance	Yes
Customizable Watch Faces	Yes
Unique Health Features	No
Indian Payment System Integration	No
Product 3	TechTime Academia
Battery Life	70%
Design & Aesthetics	Yes
Localized Language Support	Yes
Seamless Syncing	No
Water Resistance	Yes
Customizable Watch Faces	No
Unique Health Features	No
Indian Payment System Integration	No
Product Launch	TechTime Linguist
Execute	
Price, INR	23,000
Advertising, INR Million	300
Warranty Period	6 months
Distributor/Wholesaler, INR Million	100
E-commerce, INR Million	300
Telecom Partnerships, INR Million	100
Direct-to-consumer, INR Million	100

Decision Making Rationale

Product Overview and Target Segment

TechTime Linguist is a wearable device with features specifically designed for the Indian market, including multi-language support, instant translation, and culturally relevant reminders. It targets multilingual professionals and seniors who prefer interacting in regional languages. With 50% of potential users indicating a preference for localized features, TechTime Linguist addresses an important gap not fully covered by competitors like Garmin Venu, which holds a significant market share of 10% and generates substantial revenue in India. This product differentiation based on language support and cultural relevance is central to positioning TechTime Linguist as a unique option.

Product Development Choices in Prototype Phase

Battery Life (80% Capacity):

A balance was struck between functionality and power efficiency, with an 80% battery capacity enabling reasonable usage without frequent recharging, which is critical for convenience. Extensive battery life aligns well with the preferences of multilingual professionals and seniors, who may not be inclined to charge devices frequently. By offering battery life that supports prolonged usage, TechTime Linguist enhances user convenience without compromising on core features. This feature also competes favorably with alternatives, ensuring the product remains attractive to users who prioritize practicality.

Core Features (Water Resistance, Customizable Watch Faces, Health Monitoring):

These features were chosen to appeal broadly to the target demographic. Water resistance enhances durability, customizable watch faces add a personal touch, and health-monitoring features are increasingly in demand across age groups, particularly among seniors who may use the watch for basic health tracking. These features add functional value to the product, expanding its appeal and increasing its competitiveness. Offering health tracking at a competitive price point also provides an edge over premium competitors like Garmin Venu, allowing TechTime to differentiate itself as both a cost-effective and feature-rich alternative.

Unit Cost and Affordability

The decision to maintain a lower unit cost was aimed at ensuring affordability for the target market segment, which may be price-sensitive. By optimizing costs, the company can offer an accessible price that resonates with multilingual professionals and seniors.The lower unit cost allows for competitive pricing while still maintaining profitable margins. This pricing strategy aligns with the brand's goal of reaching a wide audience, increasing potential market share by positioning TechTime Linguist as a high-value yet affordable product.

Marketing Strategy and Budget Allocation

Advertising Budget (Mid-Sized Company Range)

Setting an advertising budget in line with mid-sized companies allows for impactful promotions without overextending financial resources. This budget range supports targeted marketing efforts while allowing room to allocate funds to other essential areas like product development and distribution. By optimizing the advertising spend, TechTime can maintain financial health while reaching the desired market segments through carefully selected channels. This budget level is likely to be sufficient for digital and traditional channels, ensuring comprehensive outreach and brand visibility.

Warranty Period (6 Months)

A six-month warranty period was chosen to balance product assurance with financial feasibility. While a longer warranty might appeal to some customers, the six-month period aligns with industry norms for similarly priced electronics. This warranty period gives customers confidence in the product's quality while managing the company's liability for potential repairs or replacements. It's an acceptable compromise for the target audience and matches expectations in this product category.

Channel Investment (Higher Allocation to E-Commerce)

A higher investment in e-commerce channels was chosen to maximize reach and accessibility for the target demographic, who are likely to shop online for electronics. E-commerce is cost-effective and can be easily scaled up as needed. This decision optimizes market penetration, particularly for tech-savvy multilingual professionals and seniors who may prefer the convenience of online shopping. With reduced overall channel spending, TechTime can improve margins while maintaining a solid distribution network. Allocating an equal budget to other channels ensures broad market coverage without over-investing, which allows for efficient resource allocation and cost control.

9. Common Pitfalls and Teaching Interventions

Pitfall 1: Feature Maximization Without Cost Consideration
Symptoms:

- Teams selecting all available features
- High attractiveness score but negative operating margin
- Unit cost significantly higher than price
- Team discussions focused on features without cost-benefit analysis

Intervention:

- Prompt with questions: "How does each feature impact your unit cost? Is there an optimal subset of features that maximizes profit?"
- Suggest analyzing the cost multiplier table to identify high-cost features

- Encourage comparison of feature benefit (attractiveness) versus cost impact
- Share a mini-case: "Team X selected all premium features but priced competitively, resulting in a -2% margin. How might they adjust their approach?"
- Theoretical reference: Discuss the concept of "feature creep" and the law of diminishing returns

Pitfall 2: Inadequate User Research Analysis
Symptoms:
- Rapid movement through empathize phase
- Feature selection disconnected from user personas
- Lower-than-expected attractiveness scores
- Team members referencing personal preferences rather than user data

Intervention:
- Ask teams to identify the top three needs for each persona
- Request evidence-based justification for feature prioritization
- Guide them back to sentiment analysis and preference ratings
- Facilitate a mini-exercise: "For each feature you've selected, identify which specific user need it addresses and which persona stated that need."
- Theoretical reference: Highlight the "empathy gap" concept and confirmation bias in product development

Pitfall 3: Price-Driven Strategy Without Value Creation
Symptoms:
- Very low pricing in attempt to gain market share
- Inadequate investment in product features
- Poor margins despite reasonable sales volume
- Team discussions focused on undercutting competitors

Intervention:
- Demonstrate relationship between attractiveness and price elasticity
- Show comparative data on value-based versus price-based strategies
- Encourage calculation of margin at different price points with current features
- Pose the question: "If you reduced your price by 20%, how much would your sales volume need to increase to maintain the same profit?"
- Theoretical reference: Discuss value-based pricing versus cost-plus pricing approaches

Pitfall 4: Imbalanced Channel Investment
Symptoms:
- Equal distribution of funds across all channels
- Over-investment in declining channels
- Lower-than-expected market share despite good attractiveness
- Lack of strategic rationale for channel decisions

Intervention:

- Direct attention to channel effectiveness trends over time
- Suggest analysis of investment multipliers by channel
- Encourage alignment of channel strategy with target segment behavior
- Facilitate a mapping exercise: "For each user persona, identify their preferred shopping channels based on their journey maps."
- Theoretical reference: Introduce the concept of channel cannibalization and complementarity

Pitfall 5: Ignoring Localization Opportunities

Symptoms:

- Minimal investment in localization features
- Focus on global features without market differentiation
- Strong competition from established global players
- Discussion of "universal" design without cultural context

Intervention:

- Highlight gap between current market offerings (20%) and consumer demand (60%)
- Show elasticity impact of localization features on attractiveness
- Discuss how localization creates defensive positioning against global giants
- Share the insight: "50% of surveyed users showed interest in localization features, yet current market offerings average only 20% localization."
- Theoretical reference: Introduce the concept of cultural adaptation in product design

Pitfall 6: Misalignment Between Product Concept and Target Segment

Symptoms:

- Selection of concepts that don't match prioritized user needs
- Inconsistent feature selection within a concept
- Lower-than-expected market performance despite good features
- Scattered approach without clear user focus

Intervention:

- Guide teams to match concepts with appropriate user personas
- Encourage coherent feature sets that tell a consistent product story
- Suggest analysis of competitor strengths in each segment
- Facilitate a positioning exercise: "Complete this statement: Our product is the only smartwatch that ___ for users who ___."
- Theoretical reference: Discuss market segmentation theory and the importance of coherent value propositions.

10. Instructor Tips and Insights

1. **Empathy Mapping Exercise**: Begin with an empathy mapping activity to help learners deeply understand the needs of multilingual professionals and seniors. Encourage them to consider language preferences, tech comfort levels, and lifestyle needs before discussing the product design choices.
2. **Persona Development**: Have participants create detailed user personas based on the target segments (e.g., a bilingual business consultant or a retired teacher). This brings clarity to design decisions and allows teams to align product features with real-world user behaviors.
3. **Challenge Assumptions**: Prompt learners to question existing market norms (e.g., short warranties or lack of language support). This encourages innovation by identifying overlooked needs and rethinking standard design trade-offs.
4. **Iterative Prototyping Reflections**: Emphasize the importance of iteration. Ask students to reflect on why an 80% battery capacity was chosen, and whether different iterations might prioritize other features.
5. **Triangle**: Use the F-D-V framework to evaluate decisions:
    a. **Desirability**: Does language support meet real user needs?
    b. **Feasibility**: Is health tracking realistic within cost constraints?
    c. **Viability**: Can a lower unit cost still support profitability?
6. **Group Critique and Feedback Loops**: After reviewing each team's design approach, facilitate group critiques where teams provide structured feedback to one another, simulating real-world design review processes.
7. **Pitch and Justify Design Decisions**: Conclude with teams pitching their design choices to a mock executive board. This cultivates the ability to justify product decisions based on user needs, market data, and business goals.
8. **Highlight Transferable Thinking**: Encourage students to identify how these design principles apply beyond industry —e.g., in healthcare, education tech, or public services—promoting broad application of design thinking mindsets.

Suggested Discussion Starters

1. How did your approach to the design thinking process evolve across simulation rounds?
2. How did your understanding of user needs change as you progressed through the simulation?
3. What were the most difficult trade-offs you encountered in the simulation?
4. How did you balance quantitative data with qualitative insights from user personas?
5. What assumptions did you make when data was ambiguous or incomplete?
6. How did you validate or challenge your initial assumptions through the process?
7. How did you position your product relative to competitors?
8. How did your team make decisions? What processes worked well or poorly?
9. What parallels do you see between this simulation and real product development challenges?
10. How might this design thinking approach need to be modified for different cultural context

General Microsimulation Questions

Q: How many rounds should we complete for optimal learning?
A: The Microsimulation is a standalone, single-scenario simulation usually run over 3 to 5 rounds, though instructors can extend it to 10 rounds. Two to three rounds are ideal for balancing learning and engagement. The first round sets a baseline, while later rounds help refine strategy and apply theory more deeply.

Q: Should teams be allowed to change their product concept between rounds?
A: Yes, this reflects real-world pivoting based on market feedback. However, encourage teams to justify changes based on data rather than simply reacting to disappointing results. This reinforces the "pivot or persevere" concept from lean startup methodology.

Q: How can I adapt this simulation to a different cultural context?
A: Instructors can modify the user personas, market data, and feature preferences to reflect local cultural values and behaviors. The core design thinking process remains the same, but the specific user needs, competitive landscape, and channel effectiveness may vary significantly across contexts.

Q: Is it better to focus on one strong product or develop multiple concepts?
A: The answer depends on resource availability and market dynamics.
In the simulation, participants can prototype up to three products but are limited to executing only one. This constraint underscores the importance of selecting and refining the most promising concept. A focused strategy is often more effective within the simulation's structure, especially in early rounds, were efficient resource use and clear decision-making drive success.

In the **real world**, while developing multiple concepts can help diversify risk and explore various market opportunities, it also requires significant resources. Startups and early-stage ventures often benefit from focusing on one strong product initially—gaining traction, validating the market, and building a foundation before considering portfolio expansion.

Q: How important is pricing relative to feature selection?
A: Both are critical and interdependent. Strong feature selection increases attractiveness, which reduces price sensitivity. The optimal approach balances value creation (features) with value capture (pricing). This reflects the value-based pricing theory, where willingness to pay increases with perceived value.

Q: Should investment be focused on current high-performing channels or emerging channels?

A: A balanced approach is typically best. Allocate sufficient resources to proven channels while strategically investing in growing channels like e-commerce, which shows increasing effectiveness over time. This balances exploitation of known opportunities with exploration of emerging options, a classic ambidexterity challenge.

Q: How should teams balance user desirability, technical feasibility, and business viability?

A: These three dimensions form the core of design thinking. Teams should start with user desirability (empathize phase), then consider technical feasibility (ideate and prototype phases), and finally ensure business viability (execute phase). However, all three dimensions should be considered throughout the process, with appropriate emphasis at each stage.

Technical Questions

Q: How exactly is the attractiveness score calculated?

A: The attractiveness score combines base product attractiveness with feature contributions, warranty impact, and localization effects. Each feature has an elasticity multiplier (1.01-1.06) that determines its contribution. The formula is: Base Attractiveness + (Feature Impact × Feature Selection) + (Warranty Multiplier × Warranty Period) + (Localization Impact × Localization %).

Q: What causes unexpected drops in market share despite increasing attractiveness?

A: This could be due to price increases that exceed the elasticity threshold for your attractiveness level, insufficient channel investment, or increased competitive intensity in that segment. Market share is calculated as: Base Market × (Attractiveness Elasticity × Attractiveness Score) × (Price Elasticity × Price Factor) × (Channel Investment Effectiveness).

Q: How do the battery life percentages translate to actual battery days?

A: The percentage represents investment in battery technology. The relationship is non-linear: 65% might yield 2.02 days based on the base battery capacity of the product concept. The actual calculation incorporates the base battery life of the concept and applies both competence and performance multipliers based on the investment percentage.

Conceptual Questions

Q: What's the relationship between design thinking and competitive strategy?

A: Design thinking informs value creation through user-centered innovation, while competitive strategy governs value capture and market positioning. The simulation integrates both by requiring decisions that balance user needs with competitive differentiation. This mirrors how organizations must both create compelling offerings (design thinking) and position them effectively in the market (competitive strategy).

Q: How does the simulation reflect real-world product development constraints?
A: Through limited budgets, feature trade-offs, elasticity curves that show diminishing returns, and market feedback that may not always align with expectations – all of which mirror actual product development challenges. The simulation also captures the inherent tensions between different stakeholder priorities and the need for iterative learning through testing.

Q: How does cultural context influence design thinking?
A: Cultural context shapes user needs, preferences, and behaviors, which in turn influence what solutions will be effective. The simulation demonstrates this through localization features and market-specific user personas. Effective design thinking requires deep cultural understanding and adaptation, not just applying universal principles without context.

Q: How does design thinking relate to other innovation methodologies?
A: Design thinking shares elements with other approaches like lean startup (build-measure-learn), agile development (iterative cycles), and jobs-to-be-done (user need focus). The simulation integrates aspects of all these methodologies, showing how they can complement each other in product development. The common thread is user-centricity and iterative learning.

Real-World Application Activity
To strengthen transfer of learning from the simulation to real-world contexts, conclude with this structured application exercise:

1. **Individual Reflection** (5 minutes)
    - Identify a challenge in your organization or studies that could benefit from design thinking
    - Note which specific aspects of the challenge make it suitable for this approach
    - Consider which stakeholders would need to be involved

2. **Design Thinking Plan Development** (15 minutes)
    - Create a one-page plan applying the design thinking process to your challenge:
        - Empathize: How would you gather user insights? Who are the key users?
        - Define: What might be the core problem statement?
        - Ideate: What approaches would you use to generate diverse solutions?
        - Prototype: How would you create testable versions of your solutions?
        - Execute: What implementation considerations would be critical?
    - Include specific methods and tools you would use at each stage
    - Identify potential challenges and how you would address them

3. **Small Group Feedback** (10 minutes)
    - Share your plan with 2-3 other participants
    - Provide constructive feedback on each other's plans
    - Identify strengths and opportunities for improvement
    - Suggest additional considerations or approaches

4. **Commitment to Action** (Optional)
   - Document one specific step you will take within the next week to apply design thinking
   - Set a timeline for implementing your plan
   - Identify a "learning partner" from the course to check in with on progress

This activity reinforces the practical application of design thinking concepts and increases the likelihood of knowledge transfer to real-world contexts.

Conclusion

The Design Thinking microsimulation offers a comprehensive platform for experiencing the iterative process of user-centered product development. By navigating through the phases of empathizing with users, defining problems, ideating solutions, prototyping concepts, and executing market strategies, participants develop a holistic understanding of how successful products are created.

Key takeaways include:
1. The critical importance of deep user understanding before solution development
2. The value of data-driven decision making throughout the design process
3. The necessity of balancing competing priorities: user needs, technical feasibility, and business viability
4. The power of localization and market-specific features as differentiation strategies
5. The systems nature of product development, where decisions in one area impact outcomes in others
6. The importance of cross-functional collaboration and diverse perspectives
7. The iterative nature of innovation and the value of learning from feedback

These lessons transcend the simulation context and apply broadly across industries, product categories, and cultural contexts. By mastering the design thinking methodology illustrated in this simulation, participants gain valuable skills for innovation and product development in their future careers.